Payoff Matrix with Decision Criteria

A decision matrix is a widely used tool to structure/define choices in situations of uncertainty, that is, when there are different possible alternatives, but the final outcome depends on future scenarios that are not yet fully known or controllable. According to the classical logic of decision theory, this type of matrix organizes the decision alternatives in the rows and the states of nature in the columns, which allows the expected results of each alternative to be compared in each possible scenario.

In the context of the project developed for Bayer in the Crop Science area, the decision matrix is important because agribusiness involves decisions that must be made under several uncertainties, especially due to climatic factors, for example. The rural producer needs to deal with frequent decisions, such as the best moment to plant, spray, or optimize the operation, while facing uncertainties related to climate, humidity, wind, rain, soil, or machine availability in the coming days. Therefore, the matrix helps transform a complex decision into a clearer analysis by comparing different MVP alternatives across possible future scenarios.

The importance of this matrix for the project lies precisely in showing that our project is not merely a statistical dashboard for data visualization, but rather a decision-support tool, in which it is possible to observe outputs of variables through the manipulation of metrics and the definition of assumptions, in order to recommend the best decision, which ultimately remains up to the client. With this, it becomes possible to organize alternatives and scenarios in a matrix, as well as to justify more rationally what should be prioritized in the MVP, considering not only the best possible result, but also the risk, the worst-case scenario, potential regret, and the average performance of each alternative.

In this way, it is clear that the decision matrix contributes directly to the central objective of the project: transforming fragmented data on climate, soil, operations, planting, and spraying into practical, explainable, and actionable recommendations for the producer. It allows the team to evaluate which functionality generates the most value under different conditions and helps defend the final choice based on analytical criteria, rather than intuition alone. Thus, the use of the matrix strengthens Agro Choice’s proposal as a solution capable of supporting agricultural decisions in uncertain, volatile, and highly data-dependent environments.

Output Variables	**Input Metrics**	**External Assumptions**
CV % Plantability	Seed quantity	Weather
Plasticity	Crop type	
Rainfall Scenarios	Location	
Yield	Season/Harvest Period	
	Hectares	
	Soil profile	
	Planting Date	

The variables are the results obtained from the combination of input metrics and external assumptions. Input metrics are auditable elements that can be inserted or adjusted by the client in the MVP, while external assumptions represent conditions that cannot be directly controlled by the producer, such as weather and rainfall or drought scenarios.

Regarding the input metrics, the model considers the elements that can be selected or adjusted according to the client's context and objectives. These include seed quantity, crop type, selected location, season or harvest period, hectares, planting date and soil profile that includes soil type, soil drainage, and soil pH. These inputs are essential for the simulator to estimate the expected performance of each planting strategy and calculate the payoff for each scenario.

The output variables are the results generated from the interaction between these inputs and the external conditions. Among them, CV% plantability is one of the most relevant efficiency indicators for the project, as it helps evaluate the eficiency and quality of planting performance. Yield is also considered as a central output variable, since it measures agricultural productivity in kg/ha and serves as the payoff unit in the decision matrix.

As observed in the table, Plasticity represents the crop's modeled response to the interaction between crop type, planting density, soil profile, weather, and rainfall conditions. Therefore, plasticity helps explain how the selected crop may compensate or lose efficiency under different planting strategies and environmental scenarios.In addition, rainfall impact is considered an output because the simulator evaluates how unusual rainfall or drought conditions may affect plantability and yield.

Finally, we considered only weather as a fixed external assumption, since it is a variable that is neither under our control nor under the producer's control. In addition, because weather is difficult to estimate with high precision, it was treated as a fixed assumption in the model. With this structure, the decision matrix allows the team to visualize how the simulator can support the client's decision-making process by comparing different planting strategies under uncertain conditions.

Payoff Matrix and Calculations (Spreadsheet):

Planting Density Strategy for Soybeans & Corn — What This Analysis Is About

This notebook answers a practical question every Brazilian grain farmer faces: should I plant more densely, less densely, or stick with the standard? It uses real harvest data from nearly 90,000 fields across Brazil to build a structured decision framework called a Payoff Matrix.

The Data

Everything starts from a real dataset (harvest_summary_brazil.csv) with records from soybean and corn harvests, capturing average yield in kg/ha, total production, and grain moisture. About 50,000 records are soy; 29,000 are corn.

How Scenarios Are Built

Rather than guessing at climate conditions, the analysis lets the data speak: yields are split into three groups by percentile — the bottom third represents bad years (Pessimistic), the middle third represents normal years (Baseline), and the top third represents good years (Optimistic). The average yield of each group becomes the anchor for that scenario.

Why Three Density Options, and Why ±20%?

The ±20% range around standard density is the bracket most commonly used in Brazilian field-trial research (Embrapa, ESALQ-USP) — wide enough to matter agronomically, without being unrealistic. The three alternatives are standard density, reduced density (−20% fewer plants), and increased density (+20% more plants).

The Core Insight — Why the Adjustments Are Asymmetric

This is the most important part of the model. Density doesn't affect yield the same way in good and bad years:

Reduced density (fewer plants) acts as a risk buffer. In bad years, each plant faces less competition for water and nutrients, compensating with more pods and heavier grains — a slight yield gain of +5% over standard. In good years, however, every missing plant is wasted productive potential, so there's a −8% penalty.

Increased density (more plants) is the opposite: a bet on favorable conditions. When everything goes well, a denser stand fully exploits available light and nutrients, delivering +10% over standard. But under stress, those extra plants compete for the same scarce resources and the stand can collapse — the worst outcome in the entire matrix, at −10%.

The Decision Framework

Once the nine yield values (3 densities × 3 scenarios) are in place, four classic decision criteria are applied:

Maximax — for the bold farmer chasing the best possible outcome → Density +20%
Maximin — for the cautious farmer protecting against the worst → Density −20%
Laplace — treating all scenarios as equally likely → Density +20%
Savage — minimizing future regret → Standard Density

The Bottom Line

The results are identical for soybeans and corn. The right planting density depends entirely on the farmer's risk profile, not on the crop. Farmers who want to maximize upside should plant denser; farmers who want a safety net should plant lighter. The standard density sits in the middle — never the best, never the worst, and the choice that minimizes regret.

[Corn - Spreadsheet]

[Soybean - Spreadsheet]

[Python file - Data extraction]

Conclusion and Recommendation

Decision analyzed

The decision examined in this analysis is the choice of the most appropriate planting density strategy under uncertain agronomic and environmental conditions, applied to two crops central to Brazilian agriculture: soybeans and corn. In both cases, the grower must commit to a density before the season begins, at a point when the actual climate and soil conditions of that cycle are still unknown. Three alternatives were structured for each crop — Standard Density, Reduced Density (−20%), and Increased Density (+20%) — evaluated across three possible states of the world: a Pessimistic scenario, a Baseline scenario, and an Optimistic scenario. The payoff unit throughout the analysis is crop yield measured in kg/ha.

Payoff justification

Soybeans. The Standard Density values were derived directly from observed average yields within each tercile of the dataset, based on 50,012 soybean fields from the 2025/2026 Brazilian harvest seasons. The Pessimistic scenario corresponds to the lower tercile, with an average of 4,356 kg/ha; the Baseline to the middle tercile, at 5,081 kg/ha; and the Optimistic to the upper tercile, at 5,822 kg/ha. These values are empirically grounded and serve as the reference baseline for the modeled alternatives.

The Reduced Density (−20%) values were estimated through agronomic adjustments of ±5–10% over the Standard baseline. In the Pessimistic scenario, this alternative outperforms Standard Density at 4,570 kg/ha, because lower plant populations reduce intraspecific competition for water, nutrients, and light, which increases resilience and phenotypic plasticity under stress conditions. In the Baseline and Optimistic scenarios, however, Reduced Density not only holds its ground but actually outperforms both other alternatives — reaching 5,180 and 6,400 kg/ha respectively. This pattern reflects a key agronomic property of soybean: high compensatory plasticity, meaning that under lower density, individual plants tend to branch more, produce more pods per plant, and largely compensate for the reduced stand — a behavior that is especially pronounced under favorable conditions.

The Increased Density (+20%) consistently underperforms across all scenarios for soybeans — 3,920 kg/ha in the Pessimistic, 4,930 in the Baseline, and 5,360 in the Optimistic — suggesting that beyond a certain population threshold, additional plants generate diminishing returns even under favorable conditions, while intensifying competition under stress.

Corn. The same logic applies to corn, which was derived from 29,075 fields in the same dataset. Standard Density tercile means are 5,064, 5,790, and 6,526 kg/ha for Pessimistic, Baseline, and Optimistic scenarios, respectively. As with soybeans, Reduced Density (−20%) outperforms Standard Density across all three scenarios — 5,320, 5,910, and 7,180 kg/ha — while Increased Density (+20%) consistently underperforms, with 4,560, 5,620, and 6,000 kg/ha. The direction of the results is consistent across both crops, reinforcing the robustness of the modeled pattern.

It is important to note that the ±20% alternatives are modeled estimates based on agronomic reasoning and ±5–10% adjustments, not directly measured field-trial results. They should be replaced with controlled experimental data as it becomes available.

Interpretation of the decision criteria

Four classical decision criteria under uncertainty were applied to both matrices. The table below summarizes the results for each crop:

Soybeans:

Criterion	Best Alternative	Value (kg/ha)	Decision-maker profile
Maximax	Reduced Density -20%	6,400	Optimistic – pursues highest productive ceiling
Maximin	Reduced Density -20%	4,570	Conservative — protects against worst-case climate
Minimax Regret (Savage)	Reduced Density -20%	250	Pragmatic — minimizes future regret
Laplace	Reduced Density -20%	5,383	Neutral — no scenario bias

Corn:

Criterion	Best Alternative	Value (kg/ha)	Decision profile
Maximax	Reduced Density -20%	7,180	Optimistic – pursues highest productive ceiling
Maximin	Reduced Density -20%	5,320	Conservative — protects against worst-case climate

Minimax Regret (Savage)	Reduced Density -20%	290	Pragmatic — minimizes future regret
Laplace	Reduced Density -20%	6,137	Neutral — no scenario bias

In both crops, all four criteria converge unanimously on Reduced Density (−20%) as the preferred alternative. This level of agreement across criteria is analytically significant: it means the recommendation does not depend on which decision-maker profile is assumed. Whether the grower is optimistic, conservative, neutral, or regret-averse, the analysis points to the same choice. The regret matrix further reinforces this — Reduced Density carries zero regret in the Pessimistic and Baseline scenarios for both crops, and the lowest maximum regret overall (250 kg/ha for soybeans, 290 kg/ha for corn), compared to 650 and 1,040 for Standard and Increased Density in soybeans, and 760 and 1,180 in corn.

This convergence also reflects the agronomic behavior of both crops under reduced density: the loss from fewer plants is largely offset by greater individual plant compensation, while the reduction in intraspecific competition provides a meaningful buffer against adverse conditions — making the alternative robust rather than merely optimistic.

Final recommendation

Given the full picture of the analysis, the recommendation for both soybeans and corn is Reduced Density (−20%). This conclusion is notably stronger than what would typically result from a payoff matrix analysis, because it is not driven by a single criterion or a particular risk profile — it is the dominant alternative across all four criteria simultaneously for both crops.

The recommendation is particularly well-suited for the Bayer Decision Agriculture context because it combines three qualities that matter in practice: it performs better than Standard Density in every scenario, not just in favorable ones; it carries the lowest regret under any outcome, making it the easiest recommendation to defend after the fact; and it is consistent across two different crops and three different yield scenarios derived from nearly 80,000 real field observations, which gives it a degree of empirical credibility that purely modeled alternatives cannot claim.

It should be noted that the absolute payoff advantage of Reduced Density over Standard Density varies by scenario — it is more pronounced in Optimistic conditions and narrower in the Pessimistic scenario. This means the recommendation is robust, but its magnitude is context-dependent. Growers operating in regions with frequent adverse seasons may find the Pessimistic scenario more representative, and should weigh the narrower margin accordingly. In all cases, the recommendation favors Reduced Density not because it maximizes a single metric, but because it performs well across the full range of uncertainty the grower faces.

Limitations

Several important limitations should be considered when interpreting this analysis. The payoff values for Reduced and Increased Density were estimated through modeled agronomic adjustments of ±5–10% over the Standard baseline, and have not been validated through controlled field trials; they represent informed estimates grounded in agronomic reasoning, not measured outcomes. The three scenarios were defined by equal-sized terciles of the historical yield distribution, which implicitly treats all three states of the world as equally likely — an assumption that may not hold across all regions, soil types, or climate years. The analysis also uses yield in kg/ha as the sole payoff metric, excluding variables such as input costs, seed price differentials at different densities, commodity prices, and operational constraints, which a more complete economic model would need to incorporate. Finally, the modeled relationship between density adjustments and yield response may vary significantly depending on the specific cultivar, local soil conditions, and planting date — factors that a region- or farm-level calibration would need to account for. As the project evolves and field validation data becomes available, updating the payoff values will be essential to confirm and refine this recommendation.